CRITICAL CARE MEDICINE · MIDTERM · LECTURE 2 · QUESTION BANK

ECG Monitoring

مراقبة رسم القلب — بنك الأسئلة

11	25	17
Past exam questions	NCLEX-style	Flashcards

المصادر اللي اتجمعت منها أسئلة المحاضرة دي:

كويزات سابقة (طب حالات حرجة) · أسئلة سابقة

جنب كل سؤال مصدره. لو السؤال اتكرر في كذا امتحان مكتوب كل مرة بمصدرها، وعليه علامة "تكرر". الإجابات كلها في آخر الملف.

اللَّهُمَّ لا سَهْلَ إِلَّا ما جَعَلْتَهُ سَهْلًا، وأنتَ تَجْعَلُ الحَزْنَ إذا شِئْتَ سَهْلًا — صلِّ على النبي ﷺ

Part A · Past Exam Questions

أسئلة الدفعات السابقة

1 Ventricular Tachycardia is a shockable rhythm. — كويزات سابقة (طب حالات حرجة)

a. True **b.** False

2 Electrocardiography (ECG) is considered as a diagnostic tool for STEMI. — كويزات سابقة (طب حالات حرجة)

a. True **b.** False

3 The following rhythm is considered shockable rhythm during cardiac arrest: — كويزات سابقة (طب حالات حرجة)

a. Asystole
b. Ventricular Fibrillation
c. Pulseless electrical activity (PEA)
d. Premature Ventricular Contraction (PVC)

4 In advanced cardiac life support, shockable rhythms include: — كويزات سابقة (طب حالات حرجة)

a. Ventricular tachycardia in hemodynamically stable patient
b. Pulseless ventricular tachycardia
c. Premature ventricular tachycardia
d. B and C

5 Pulseless ventricular tachycardia is considered from non-shockable rhythms during advanced cardiac life support. — كويزات سابقة (طب حالات حرجة)

a. True **b.** False

6 What is a typical finding on an ECG in a patient with PE? — أسئلة سابقة

a. Left bundle branch block **b.** Sinus tachycardia
c. Normal sinus rhythm **d.** ST-segment elevation

7 Which of the following investigations can identify arrhythmias that could lead to embolic stroke? أسئلة سابقة

a. CT scan
b. Duplex Ultrasound
c. Electrocardiogram (ECG)
d. MRI

8 Which rhythm is considered shockable in ACLS? أسئلة سابقة

a. Asystole
b. Pulseless Electrical Activity (PEA)
c. Ventricular Fibrillation (VF)
d. Normal sinus rhythm

9 What is the key difference between shockable and non-shockable rhythms in ACLS? أسئلة سابقة

a. Shockable rhythms require defibrillation, non-shockable rhythms do not
b. Shockable rhythms are always treated with epinephrine
c. Non-shockable rhythms respond to defibrillation
d. Shockable rhythms require higher doses of epinephrine

10 What is the first action to take if a patient shows signs of a shockable rhythm like VF or VT? أسئلة سابقة

a. Administer epinephrine
b. Perform CPR immediately
c. Perform defibrillation
d. Insert an advanced airway

11 When is defibrillation indicated in a patient with a non-shockable rhythm like PEA or Asystole? أسئلة سابقة

a. Immediately after CPR
b. After 1 minute of CPR
c. Never
d. Only after the administration of epinephrine

Part B · NCLEX-Style Practice

أسئلة على طريقة NCLEX (إضافة)

N1 On standard ECG paper, one small square equals: NCLEX

a. 0.04 seconds
b. 0.2 seconds
c. 1 second
d. 0.12 seconds

N2 The P wave on the ECG represents: NCLEX

a. Ventricular depolarization
b. Atrial depolarization
c. Ventricular repolarization
d. Atrial repolarization

N3 The normal PR interval is: NCLEX

a. 0.04–0.08 s
b. 0.12–0.20 s
c. 0.24–0.40 s
d. > 0.20 s

N4 The normal QRS duration is: NCLEX

a. < 0.12 seconds
b. 0.12–0.20 seconds
c. > 0.20 seconds
d. Exactly 0.40 seconds

N5 Using the 1500 method, if there are 15 small squares between two R waves, the heart rate is: NCLEX

a. 60 bpm
b. 100 bpm
c. 150 bpm
d. 75 bpm

N6 For an irregular rhythm, the best way to calculate heart rate is: NCLEX

a. 1500 method
b. 300 method
c. Count QRS complexes in a 6-second strip × 10
d. Measure the PR interval

N7 Correct placement of the V1 chest lead is: NCLEX

a. 4th intercostal space, right sternal border
b. 5th intercostal space, midclavicular line
c. 4th intercostal space, left sternal border
d. Left anterior axillary line

N8 Steps of systematic rhythm interpretation include (select all): SELECT ALL

a. Determine the rate
b. Assess regularity
c. Examine P waves
d. Measure PR and QRS
e. Check the patient's blood sugar

N9 Sinus bradycardia is defined as a sinus rhythm with a rate: NCLEX

a. < 60 bpm
b. < 80 bpm
c. > 100 bpm
d. < 40 bpm only

N10 An ECG shows an irregularly irregular rhythm with no P waves and fibrillatory baseline. This is: NCLEX

a. Atrial flutter
b. Atrial fibrillation
c. Sinus arrhythmia
d. Third-degree block

N11 A 'saw-tooth' pattern of flutter waves is characteristic of: NCLEX

a. Atrial flutter
b. Ventricular fibrillation
c. Asystole
d. First-degree AV block

N12 Wide, bizarre, early QRS complexes without a preceding P wave are: NCLEX

a. Premature atrial contractions
b. Premature ventricular contractions (PVCs)
c. Junctional rhythm
d. Sinus beats

N13 The monitor shows ventricular tachycardia. The nurse's FIRST action is: NCLEX

a. Defibrillate immediately
b. Check the patient's pulse and responsiveness
c. Give amiodarone
d. Record a 12-lead ECG

N14 Chaotic, irregular waveform with no identifiable P, QRS or T waves is: NCLEX

a. Ventricular fibrillation
b. Atrial fibrillation
c. Sinus tachycardia
d. Junctional rhythm

N15 Before defibrillating, the nurse should first: NCLEX

a. Confirm the flat line in two leads
b. Check the patient and make sure everyone is clear
c. Give atropine
d. Synchronize the shock

N16 A flat line appears on the monitor. The nurse should first: NCLEX

a. Defibrillate
b. Check the patient, leads and gain; confirm in another lead
c. Give amiodarone
d. Synchronize cardioversion

N17 Pulseless electrical activity (PEA) means: NCLEX

a. Organized electrical activity on the monitor without a palpable pulse
b. Flat line
c. VF with a pulse
d. Normal rhythm with hypotension

N18 Which rhythms are shockable? (select all) SELECT ALL

a. Ventricular fibrillation
b. Pulseless ventricular tachycardia
c. Asystole
d. PEA
e. Sinus bradycardia

N19 Third-degree (complete) heart block is recognized by: NCLEX

a. P waves and QRS complexes beating independently (AV dissociation)
b. Progressive PR prolongation then a dropped beat
c. Constant PR > 0.20 s
d. Irregularly irregular rhythm

N20 Progressive lengthening of the PR interval until a QRS is dropped describes: NCLEX

a. Mobitz I (Wenckebach) second-degree block
b. Mobitz II block
c. First-degree block
d. Atrial flutter

N21 Peaked (tall tented) T waves on the monitor suggest: NCLEX

a. Hypokalemia
b. Hyperkalemia
c. Hypocalcemia
d. Hypothermia

N22 ECG evidence of STEMI is: NCLEX

a. ST elevation in ≥ 2 contiguous leads or new LBBB
b. ST depression in one lead
c. Tall P waves
d. Prolonged PR interval

N23 Nursing responsibilities in continuous cardiac monitoring include (select all): SELECT ALL

a. Clean and dry skin before electrode placement
b. Change electrodes every 24 hours or as needed
c. Set alarm limits and never switch them off
d. Check the patient, not just the monitor
e. Silence alarms that keep ringing

N24 Synchronized cardioversion differs from defibrillation because it: NCLEX

a. Delivers the shock on the R wave
b. Is used for VF
c. Uses no energy
d. Is given without consent in stable patients

N25 Sinus tachycardia is a regular sinus rhythm with a rate of: NCLEX

a. > 100 bpm
b. 60–100 bpm
c. < 60 bpm
d. > 300 bpm

Part C · Quizlet Flashcards

كروت مراجعة سريعة

غطّي العمود الأبيض بورقة واختبر نفسك: شوف المصطلح وقول التعريف.

ECG paper	Small square 0.04 s / 1 mm; large square 0.2 s / 5 mm; 10 mm = 1 mV.	**Waves**	P = atrial depolarization · QRS = ventricular depolarization · T = ventricular repolarization.
Normal intervals	PR 0.12–0.20 s · QRS < 0.12 s · QTc < 0.44 s.	**Heart rate**	300 ÷ large squares, 1500 ÷ small squares, or QRS in 6 s × 10 (irregular).
5-step interpretation	Rate → rhythm (regularity) → P waves → PR interval → QRS width.	**Limb leads**	RA white, LA black, LL red, RL green ('white on right, smoke over fire').
Chest leads	V1 4th ICS right sternal border · V2 4th ICS left · V4 5th ICS MCL · V6 mid-axillary.	**Sinus brady / tachy**	< 60 / > 100 bpm with normal P before each QRS.

Atrial fibrillation	Irregularly irregular, no P waves; stroke risk → anticoagulation.
Atrial flutter	Saw-tooth F waves, atrial rate ~300.
PVC	Early wide bizarre QRS without P; dangerous if frequent, multifocal, R-on-T.
VT	≥ 3 wide QRS, rate > 100; pulseless → shock; with pulse unstable → synchronized cardioversion.
VF	Chaotic baseline, no output → defibrillate + CPR.
Asystole / PEA	Non-shockable → CPR + epinephrine 1 mg every 3–5 min; search Hs & Ts.
Heart blocks	1st: PR > 0.20 · Mobitz I: PR lengthens then dropped beat · Mobitz II: constant PR, dropped beats · 3rd: AV dissociation → pacing.
STEMI on ECG	ST elevation in ≥ 2 contiguous leads; II, III, aVF inferior · V1–V4 anterior · I, aVL, V5–V6 lateral.
Potassium	Hyperkalemia: peaked T, wide QRS · Hypokalemia: flat T, U waves.

Answer Key

الإجابات

1. a — True
إجابة مقترحة (المصدر مفيهوش إجابة رسمية). الطالب في الصورة اختار False، بس الصح True: الـ pulseless VT والـ VF هما الـ shockable rhythms. أما الـ VT اللي معاها نبض فبتتعالج بـ cardioversion مش defibrillation.

2. a — True
إجابة مقترحة (المصدر مفيهوش إجابة رسمية). الطالب اختار False، بس الصح True: الـ STEMI تشخيصه أصلاً بالـ ECG (ST elevation في ليدين متجاورين أو أكتر).

3. b — Ventricular Fibrillation
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

4. b — Pulseless ventricular tachycardia
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

5. b — False
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

6. b — Sinus tachycardia

7. c — Electrocardiogram (ECG)

8. c — Ventricular Fibrillation (VF)

9. a — Shockable rhythms require defibrillation, non-shockable rhythms do not

10. c — Perform defibrillation

11. c — Never

N1. a — 0.04 seconds
Large square = 0.2 s (5 small squares); 5 large squares = 1 s.

N2. b — Atrial depolarization
QRS = ventricular depolarization; T wave = ventricular repolarization.

N3. b — 0.12–0.20 s
PR > 0.20 s = first-degree AV block.

N4. a — < 0.12 seconds
A wide QRS (≥ 0.12 s) suggests a ventricular origin or bundle branch block.

N5. b — 100 bpm
1500 ÷ 15 = 100. With large squares: 300 ÷ number of large squares.

N6. c — Count QRS complexes in a 6-second strip × 10
The 6-second method works for irregular rhythms like AF.

N7. a — 4th intercostal space, right sternal border
V2 = 4th ICS left sternal border; V4 = 5th ICS midclavicular.

N8. a, b, c, d — Determine the rate; Assess regularity; Examine P waves; Measure PR and QRS
Always assess the patient too: is there a pulse?

N9. a — < 60 bpm
Treat if symptomatic: atropine 1 mg IV first-line.

N10. b — Atrial fibrillation
AF raises the risk of thrombo-embolic stroke → anticoagulation.

N11. a — Atrial flutter
Atrial rate about 300/min.

N12. b — Premature ventricular contractions (PVCs)
Frequent or multifocal PVCs, couplets or R-on-T can lead to VT/VF.

N13. b — Check the patient's pulse and responsiveness
Pulseless VT → defibrillate; VT with pulse & unstable → synchronized cardioversion; stable → amiodarone.

N14. a — Ventricular fibrillation
VF = no cardiac output → immediate defibrillation and CPR.

N15. b — Check the patient and make sure everyone is clear
Also check lead/monitor artefact: a loose lead can mimic VF or asystole.

N16. b — Check the patient, leads and gain; confirm in another lead
Asystole is non-shockable: CPR + epinephrine 1 mg every 3–5 min.

N17. a — Organized electrical activity on the monitor without a palpable pulse
Search for reversible causes (Hs and Ts).

N18. a, b — Ventricular fibrillation; Pulseless ventricular tachycardia
Asystole and PEA are treated with CPR and epinephrine.

N19. a — P waves and QRS complexes beating independently (AV dissociation)
Treatment: pacing (transcutaneous then transvenous).

N20. a — Mobitz I (Wenckebach) second-degree block
Mobitz II: constant PR with sudden dropped beats — more dangerous.

N21. b — Hyperkalemia
Hypokalemia gives flat T waves and U waves.

N22. a — ST elevation in ≥ 2 contiguous leads or new LBBB
ST elevation in II, III, aVF = inferior MI; V1–V4 = anterior MI.

N23. a, b, c, d — Clean and dry skin before electrode placement; Change electrodes every 24 hours or as needed; Set alarm limits and never switch them off; Check the patient, not just the monitor
Treat the patient, not the monitor.

N24. a — Delivers the shock on the R wave
Used for unstable tachycardias with a pulse (AF, flutter, SVT, VT with pulse).

N25. a — > 100 bpm
Treat the cause: pain, fever, hypovolemia, anxiety, hypoxia.